

Supplementary Materials (V3)

From Material Turnover to Causal Memory: Organizational Continuity in a Simulated Droplet

Autonomous research agent (see AI-authorship disclosure in the main text)

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Scope of this revision (V3). This supplement adds a complete, self-contained model specification (S1) sufficient to reimplement every reported number from scratch, and states the decoding pipeline, numerical scheme, and data-availability wording exactly. Sections S2–S10 are the provenance, registry, and reproduction material carried forward from the previous supplement, lightly renumbered.

1 S1. Complete model specification

All fields live on a periodic 64×64 square lattice, integrated by explicit forward Euler with time step $dt = 0.1$. The periodic five-point Laplacian is

$$(\Delta X)_{ij} = X_{i+1,j} + X_{i-1,j} + X_{i,j+1} + X_{i,j-1} - 4X_{ij}. \quad (1)$$

The engine is the frozen scaffold blob 7c91b91; the memory field is the frozen IOM-00 writing law (dc04f5d) with the two orthogonal readouts of MCM-00. With $\lambda_+ = \lambda_- = 0$ the base fields (ρ, U, V, c, N) evolve bit-identically to the memory-free scaffold.

S1.1 State variables

ρ (scaffold density); $U = \rho u$, $V = \rho v$ (extensive internal bistable species, concentrations u, v); c (shared attractant); N (nutrient); C (passive cohort tracer, $\sum_c C = \rho$ exactly); $M_k = \rho m_k$ for $k \in \{1, 2\}$ (extensive memory, intensive $m_k \in [-1, 1]$).

S1.2 Cohesion: chemotactic advection with volume exclusion

Per axis, for the forward neighbour ($\rho^+ = \text{roll}(\rho)$, $c^+ = \text{roll}(c)$, $dc = c^+ - c$) the face flux is upwind in dc :

$$\chi = \frac{\chi_0}{1 + \left(\frac{1}{2}(c + c^+)/c_{\text{sat}}\right)^2}, \quad J = \underbrace{\chi dc \rho_{\uparrow} \left(1 - \frac{\rho_{\downarrow}}{\rho_{\text{max}}}\right)_+}_{\text{advection}} - \underbrace{D_{\rho}(\rho^+ - \rho)}_{\text{diffusion}}, \quad (2)$$

where $\rho_{\uparrow} = \rho$, $\rho_{\downarrow} = \rho^+$ if $dc > 0$, else swapped. The density update is the flux divergence $\partial_t \rho = -\sum_{\text{axes}} (J - \text{roll}(J, 1))$. The extensive carriers U, V, C, M_k advect with the same flux using donor-cell intensive fractions $(U/\rho, V/\rho, C/\rho, M_k/\rho)$.

S1.3 Growth (nutrient uptake), death

With $\sigma = (u - v)/(u + v)$, $q = (1 - \rho/\rho_{\max})_+$, and dose readout $m_+ = \tanh(m_1 + m_2)$:

$$g = dt \, g_0 \, \rho \, N \, q \, (1 + \beta\sigma) \, (1 + \lambda_+ m_+), \quad g \leftarrow \text{clip}(g, 0, N), \quad (3)$$

$$N \leftarrow N - g, \quad \rho \leftarrow \rho + g, \quad U \leftarrow U + gu, \quad V \leftarrow V + gv, \quad M_k \leftarrow M_k + g m_k. \quad (4)$$

New mass inherits the local internal state and local memory. The uptake observable is uptake = g . Death is homogeneous, $X \leftarrow (1 - dt \, k)X$ for $X \in \{\rho, U, V, C, M_k\}$, so intensive concentrations and memory are invariant under pure turnover.

S1.4 Internal bistable identity (confined to the scaffold)

Where $\rho > 10^{-4}$ and $a > 0$, on concentrations:

$$\dot{u} = \frac{a}{1 + (v/K)^2} - u, \quad \dot{v} = \frac{a}{1 + (u/K)^2} - v, \quad u \leftarrow u + dt(\tau\dot{u} + D_{\text{int}}\Delta u), \quad (5)$$

(analogously v), then clipped to ≥ 0 and re-made extensive $U = \rho u$, $V = \rho v$. This network never enters χ , c , or any cohesive term.

S1.5 Memory writing (frozen) and the two readouts

Let uptake be the entity-mean uptake. The single local experience signal is

$$\Psi = \tanh(k_{\text{exp}}(N - c) + k_{\text{up}}(\text{uptake} - \text{uptake})). \quad (6)$$

Each intensive component, where $\rho > 10^{-4}$, obeys write / two-timescale forgetting / templating / diffusion:

$$m_k \leftarrow \text{clip}\left(m_k + dt[\eta_w \Psi - \eta_{d,k} m_k + \eta_t(\widetilde{m}_k - m_k) + D_m \Delta m_k], -1, 1\right), \quad (7)$$

with $\eta_{d,1}$ (fast) and $\eta_{d,2}$ (slow), and the templating reference $\widetilde{X} = \frac{1}{4}(X_{i\pm 1,j} + X_{i,j\pm 1})$ (periodic four-neighbour mean). Readouts:

$$\text{dose (channel 1): } m_+ = \tanh(m_1 + m_2) \rightarrow \text{uptake gain } (1 + \lambda_+ m_+) \text{ [in S1.3]}, \quad (8)$$

$$\text{order (channel 2): } m_- = \tanh(m_1 - m_2) \rightarrow c \text{ production } (1 + \lambda_- m_-). \quad (9)$$

The attractant and nutrient close the system:

$$c \leftarrow c + dt(D_c \Delta c + s \rho (1 + \lambda_- m_-) - \delta c), \quad N \leftarrow N + dt(D_N \Delta N + F(N_0 - N)). \quad (10)$$

S1.6 Frozen parameters

Scaffold (blob 7c91b91): $\chi_0 = 9.5$, $D_\rho = 0.07778$, $D_c = 0.68$, $s = 0.2$, $\delta = 0.06909$, $g_0 = 0.06154$, $k = 0.02588$, $F = 0.02421$, $D_N = 0.5$, $N_0 = 1.0$, $\rho_{\max} = c_{\text{sat}} = 1.0$, $a = 2.0$, $K = 0.5$, $D_{\text{int}} = 0.008$, $\tau = 0.20$, $\beta = 0.6$, lattice 64, $dt = 0.1$. Memory (selected candidate **C1c**, frozen **FREEZE_C1c.json**): $\eta_w = 0.015$, $\eta_{d,1} = 0.35$, $\eta_{d,2} = 0.006$, $\eta_t = 0.010$, $D_m = 0.010$, $\lambda_+ = 0.25$, $\lambda_- = 0.15$, $k_{\text{exp}} = 1.0$, $k_{\text{up}} = 1.0$, $n_{\text{comp}} = 2$. (C1c is the de-saturated writing correction of the IOM-00 default $\eta_w = 0.05$, $\eta_{d,1} = 0.03$, $k_{\text{exp}} = 2.0$.)

S1.7 History coordinates

Each history applies an early and a late environmental drive of amplitudes $a_{\text{early}}, a_{\text{late}}$. The two decoded coordinates are the cumulative dose $h_1 = a_{\text{early}} + a_{\text{late}}$ (global, non-individuating) and the temporal order / asymmetry $h_2 = a_{\text{late}} - a_{\text{early}}$.

S1.8 Entity detection

Connected components (periodic 4-connectivity) of the mask $\{\rho > 0.30 \rho_{\text{max}}\}$; components with ≥ 12 cells are entities. Centroids use the periodic circular-mean estimator; radius of gyration and all phenotype statistics are computed on the entity cells with ρ -weighting.

S1.9 Decoding pipeline (how every R^2 is obtained)

Memory feature vector per entity: for each component m_k over the entity cells, the five order statistics [mean, std, P_{10} , P_{50} , P_{90}], concatenated over $k = 1, 2$ to a **10-D** vector (entities with < 4 cells are undefined and censored). Decoder: **ridge regression**, regularisation $\lambda = 1.0$, evaluated by **grouped leave-one-history-out** cross-validation (the held-out fold is an entire history class, never a single entity, to prevent leakage). Uncertainty: donor-level **percentile bootstrap**, $n_{\text{boot}} = 3000$, resampling the grouping unit (history/seed), reported as 95% CIs. Longitudinal tracking: periodic mask-overlap matching, overlap threshold $\theta = 0.1$, evaluated **every 5 steps**, censored on track loss; the tracker never reads h_1, h_2, M , or any label. Turnover fraction M is the surviving mass fraction of the original cohort field.

2 S2. Protocol history (the nine-experiment ladder)

Each experiment below was specified by a protocol committed *before* its prospective family was generated and executed once (see the provenance table in the main text for protocol / commit / freeze / execution timestamps and hashes). HMC (material continuity): snapshot nearly sufficient, no strong history-specific hidden persistence. HSI (hidden state): generic causal attractor, not individuating (AUC 0.3–0.4). IOM-00: designed two-component memory; causal, erasable, transplantable; readout ≈ 1 -D. MCM-00: $m_- \rightarrow$ attractant makes order readable ($\sim 70/73 \times$ ablation collapse). WD-01 Phase B: grouped-evaluation correction ($0.57 \rightarrow 0.19$); rank ≈ 1 . Phase C: de-saturated writing (C1c); two storage coordinates at rest, causal response ≈ 1 -D. SMC-01: h_2 frame-free dispersion; a turnover collapse later corrected. DMM-01: correction — inter-family variance, not smoothing/dilution; underpowered. H2-CERT-01: sealed kill-switch, h_2 point 0.34 at $M \approx 0.19$ with a CI spanning the 0.50 threshold (retention *not established*). TCA-01: tracker-continuity audit; longitudinal re-certification of h_1 .

3 S3. Branch / commit registry

experiment	result commit
HMC	25c419a
HSI	709f963
IOM-00	0ea1250
MCM-00	5841b9b

WD-01 Phase B	9a39f8c
WD-01 Phase C (seal / result)	be08738 / a07b95b
SMC-01 (seal / result)	d0f47b6 / 3dc8c35
DMM-01 (seal / result)	5bd46e7 / 49e96ad
H2-CERT-01 (seal / result)	5e1196f / 1f8f789
consolidation / observer	112aace / 6891ef8
tracker audit / post-incident	568af39 / 7afd6e4
short-paper V0 / full manuscript	a97909a / (this commit)

Main branch never modified (HEAD f3921a4); nothing pushed (origin/main e17f431).

4 S4. Sealed prospective family hashes (SHA-256, generated before selection, executed once)

experiment	seeds (dev / prospective)	SHA-256
Phase C	34001-3 / 35001-3	c6d0cd3c5b82aa122ac7e5649888fbc53ca6faa8db68e2c1b48863e87fb9e202
SMC-01	36001-3 / 36501-3	2d6986fe2499cb52fdc309f5b1867cff56cb04b4bd7884633c4c2a148d5792ef
DMM-01	37001-3 / 37501-3	923d8890ea25ab5a136078486f282ad8e07f4cdbaa125da40ef930f5bb8a00b6
H2-CERT	— / 38501-4	4265b98cacb705a7bea56a2e0a5bbe36e10fec3d977e7da6e655037189193fee

Seed families disjoint across experiments (32xxx-38xxx). “Executed once” means the prospective family was run a single time after the protocol freeze; no prospective seed was re-run to obtain a preferred number.

5 S5. Parameter grids and negative candidates

Writing-redesign candidates compared at development (Phase C): C0 (frozen baseline), C1a/C1b/C1c (dynamic-range corrections varying $\eta_w, \eta_{d,1}, \eta_{d,2}, k_{\text{exp}}$), C2a (distinct write signals). Development $\min(R^2(h_1), R^2(h_2))$ on family F_mid: C0 0.19, C1a 0.84, C1b 0.61, C1c 0.84, C2a 0.79; C1c selected for the highest stored second-dimension ratio ($\sigma_2/\sigma_1 = 0.229$) at equal top score, by minimality (a same-signal C1). Maintenance candidates (DMM Phase A, diagnostic only): templating and diffusion ablations and static memory relaxation all left the h_2 decode essentially unchanged. Negative variants were retained, not discarded.

6 S6. Gate certificates (machine-readable)

JSON gate certificates are committed per experiment: `results/wd01/wd01_gate_certificate.json`, `results/wd01_phasesec/wd01_phasesec_gate_certificate.json`, `results/smc01/smc01_gate_certificate.json`, `results/dmm01/dmm01_gate_certificate.json`, `results/h2cert/h2cert_gate_certificate.json`. Each records the preregistered gates and the observed values, including the gates that were *not* met.

7 S7. Tracker unit tests (TCA-01)

(i) Selection audit: the historical convenience rule is $\arg \max$ size, reselected per frame; five hard switches occurred on seed 38501. (ii) Periodic translation invariance: whole-world shifts (including boundary-straddling) leave size and memory statistics bit-invariant. (iii) Sensitivity: h_1 decodes 0.96/0.98/0.99 from largest / second-largest / longitudinal trackers (tracker-independent). (iv) Hold-out: untouched seeds 38502–38504, 36/36 track survival, h_1 deep $R^2 = 0.98$ CI [0.97, 1.00], h_2 point 0.34. Specification: docs/audit/TCA_01_TRACKER_SPEC.md.

8 S8. Seed-family results (h_1 turnover) and the re-audited intervals

DMM pool (4 seeds): h_1 0.98 $\rightarrow$ 0.99 $\rightarrow$ 0.99 over $M = 1.00 \rightarrow 0.58 \rightarrow 0.36$. H2-CERT sealed (4 seeds): 0.94 $\rightarrow$ 0.93 $\rightarrow$ 0.93 over $M = 1.00 \rightarrow 0.44 \rightarrow 0.21$. Longitudinal hold-out (untouched seeds): initial $R^2 = 0.92$ CI [0.86, 0.99], deep 0.98 CI [0.97, 1.00] — the paper’s primary quantitative result. Re-audited comparators (grouped bootstrap, $n_{\text{boot}} = 3000$): transplant h_1 0.61 CI [−0.69, 0.87]; in-place h_1 0.50 CI [−1.39, 0.92]; memory-inert / erased deterministic −0.19; memory vs. size+mass 0.93 vs. 0.64 (CIs overlap); longitudinal h_2 deep point 0.34 CI [−0.89, 0.87] (spans 0.50; *not established*). Full methodology: STATISTICAL_REAUDIT.md.

9 S9. Reproduction commands and data availability

Environment: python3, numpy, scipy, matplotlib; PYTHONPATH=\$REPO:\$REPO/results/wd01_phasec. Per experiment, resumable runners are committed beside their data: results/wd01/run_diag.py; results/wd01_phasec/{dev_runner,prosp_runner,causal_runner,causal_inplace}.py; results/smc01/{fig1_model_ladder,fig2_h1_causal_v3,fig3_tracked_3panel,fig3b_longitudinal_certification,fig4_order_m_minus,fig5_phasec_figure,fig6_h2cert_figure,fig7_paper_claim_ladder}.py; results/dmm01/{phaseA,pooled,prosp}.py; results/h2cert/{pilot,cert}.py; the tracker hold-out reruns deterministically from the H2-CERT manifest seeds 38502–38504. Viewer: scripts/observe_droplet.py. **--preset story.** *Data availability.* All raw arrays, JSON certificates, sealed manifests, and figure generators referenced here are committed to the branch paper/organizational-memory-boundaries-01 of the repository emergent-dynamics-lab; every reported number derives from a file in that commit and no analysed run was excluded from the reported statistics.

10 S10. Raw-data and figure registry

Raw: results/{sc_mcm,wd01,wd01_phasec,smc01,dmm01,h2cert,observer}/*.pkl,*.json. Figures (paper): Fig. 1 fig1_model_ladder; Fig. 2 fig2_h1_causal_v3 (predicted-vs-observed with bootstrap CI); Fig. 3 tracked_3panel; Fig. 3b fig3b_longitudinal_certification (primary result); Fig. 4 fig4_order_m_minus; Fig. 5 phasec_figure; Fig. 6 h2cert_figure; Fig. 7 paper_claim_ladder (corrected historical summary). Each figure traces to a committed generator script.